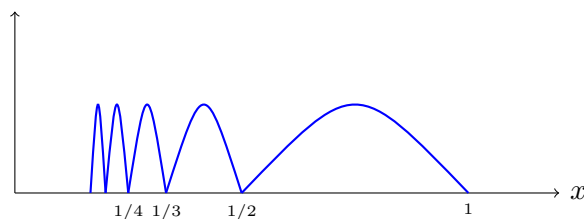


Explainer: Problem 7 (Rudin 7.5) — Disjoint Bumps: Absolute but Not Uniform Convergence

The sequence

$$f_n(x) = \begin{cases} \sin^2(\pi/x) & \text{if } \frac{1}{n+1} \leq x \leq \frac{1}{n}, \\ 0 & \text{otherwise.} \end{cases}$$

Each f_n is a bump supported on the tiny interval $[\frac{1}{n+1}, \frac{1}{n}]$.



Each bump f_n lives on $[1/(n+1), 1/n]$ and peaks at 1.

Two claims

1. $\{f_n\}$ converges pointwise to a **continuous** function (it's the zero function), but not uniformly.
2. The series $\sum f_n$ converges absolutely for every x , but not uniformly.

Why pointwise $\rightarrow 0$

Fix any $x \in \mathbb{R}$.

- If $x \leq 0$: every $f_n(x) = 0$.
- If $x > 1$: also 0 for all $n \geq 1$.
- If $0 < x \leq 1$: x lies in at most one interval $[1/(n+1), 1/n]$, so $f_n(x) \neq 0$ for only *one* value of n ; for all larger n , $f_n(x) = 0$.

So $f_n(x) \rightarrow 0$ for every x . The limit function is identically 0 — continuous.

Why NOT uniformly

The bumps all reach height 1 (take x where $\pi/x = \pi/2$, i.e., $x = 2$... wait, you need this inside the bump support). In each bump interval $[1/(n+1), 1/n]$, does $\sin^2(\pi/x)$ attain the value 1?

Yes: $\sin^2(\pi/x) = 1$ when $\pi/x = \pi/2 + k\pi$, i.e., $x = 2/(2k+1)$. For each n , you can find such a k making x lie in $[1/(n+1), 1/n]$ (since the interval is large enough to contain one of these points for big n).

So $\sup_x f_n(x) = 1$ for every n , which doesn't go to 0. Not uniform.

The series $\sum f_n$

At each x , at most *one* $f_n(x) \neq 0$. So the series is essentially a single term at every x — **absolutely convergent** (trivially).

But not uniformly: The partial sum $S_N(x) = \sum_{n=1}^N f_n(x)$ differs from the full sum by $\sum_{n>N} f_n$, which is *nonzero* on $(0, 1/(N+1)]$ and still reaches height 1. So $\sup |S - S_N|$ doesn't $\rightarrow 0$.

Takeaway

Absolute pointwise convergence is much weaker than uniform convergence. Even when each x sees only one nonzero term (so absolute convergence is automatic), the *location* of the nonzero term can shift, keeping the worst-case error bounded away from 0.